

DAY 02

Network Basics

Cybersecurity Learning Journey

Introduction

Hi this day 02 of learning cybersecurity.

Network and Network Security

A network in cybersecurity is a connected group of computers, servers, and devices that share data, while network security is the specific practice of protecting that communication pathway and its data from cyber threats

What is a Network?

Digital Highway: A system linking devices (like laptops, phones, and printers) via cables or Wi-Fi so they can talk to each other.

Core Parts: Includes end devices, routers (which direct traffic between different networks), and switches (which connect devices inside a single network).

Rules (Protocols): Standards like TCP/IP and HTTP control how data moves safely from one point to another.

OSI Model

OSI model stands for "Open Systems Interconnection". It is the main parts of network devices for communicate across the devices

There are 7 parts

7. Application :- Where humans process data/information.
 6. Presentation :- Ensure the data in a usable format
 5. Session :- Capable for maintaining connections
 4. Transport :- ensure end-to-end delivery from sender to receiver.
 3. Network :- Responsible for which path packets should travel on a network
 2. Data Link :- Responsible for which physical devices packets should go to
 1. Physical :- The physical infrastructure to transport data
- // First 3 layers (Application, Presentation, Session) are associated with softwares.
- // Last 3 layers (Network, Data Link, Physical) are associated with Hardware
- // And 4th Layer Transport transport enusres the data packets to correct destination

IP Addressing

IP (Internet Protocol)

IP is used to communicate across networks, not just across physical links, but between networks of routers. The addressing scheme in use is either IPv4 ("IP Version 4") or IPv6 ("IP Version 6")

IPv4: Uses a 32-bit number system written in decimal format with four octets (e.g., 192.168.1.1). It supports about 4.3 billion unique addresses.

IPv6: Uses a 128-bit alphanumeric system written in hexadecimal format with colons (e.g., 2001:db8::ff00:42:8329). It was created to replace IPv4 as addresses run out. You can learn more about technical structures through the ManageEngine IP Address Guide.

IP networks can be broken into different sections, often called subnets

This is accomplished by adding an extra piece of information, together with the IP address, called a netmask

MAC Address vs IP Address

What is the difference between MAC and IP?

Q1> What is the difference between mac and ip?

MAC Address and IP Address are used to uniquely define a device on the internet. NIC Card Manufacturer provides the MAC Address, on the other hand, the Internet Service Provider provides the IP Address

What is MAC Address?

MAC, or Media Access Control is a unique identifier that is shaped on the computer hardware to connect the system to other systems of the network address. It uses the Network Interface Controller to create a communication segment to establish communication between systems on the same network. MAC Address works for a local network. The MAC address is used by the Media Access Control (MAC) sublayer of the Data-Link Layer. MAC Address is worldwide unique since millions of network devices exist and we need to uniquely identify each

MAC Address is represented as a group of 6 hexadecimal numbers, each group consisting of two hexadecimal numbers. It is also called a 6-byte hexadecimal number.

Difference Between MAC Address and IP Address

MAC addresses are hardware identifiers, while IP addresses represent network locations. Together, they ensure proper data routing on the internet.

The main difference between MAC and IP address is that MAC Address is used to ensure the physical address of the computer. It uniquely identifies the devices on a network. While IP addresses are used to uniquely identifies the connection of the network with that device takes part in a network.

Conclusion

Conclusion

Both MAC address & IP address are unique identifiers that connect to those systems which have an internet connectivity. They differ in the aspect that MAC address is a local area network address, whereas IP address is a global area network. There are many differences too in terms of their configurations, format which once read in the tabular format shown above would help us to understand the differences and benefits clearly.

Routing & Local Network Communication

Default Gateway in Networking

Default Gateway in Networking

Routing is the most important process in networking. The process of routing includes various networking devices for packet forwarding one of which is the default gateway. The default gateway forwards the packets from the client to other network

Default Gateway:

The default gateway is the node that forwards the packet from the source to other networks when there is no routing information about the destination i.e. host (or router) does not know where the destination is present.

A default gateway is a route to which information is passed when the device does not know where the destination is present.

It is used when there is no routing information available about the destination.

It is a node that allows the communication of computers on different networks.

'Default' here means the default route which is to be taken when the host does not know where the destination is.

It is most commonly used for webpage access.

This is an important part of networking for routing the data and finding the corresponding destination which is in another network.

Explain ARP

Explain ARP?

The Address Resolution Protocol (ARP) is a communication protocol used to map a logical IPv4 address to a physical hardware Media Access Control (MAC) address on a local area network (LAN)

How ARP Works

ARP Request: When a device wants to talk to another device on the same network, it checks its local memory first. If it does not know the target physical address, it broadcasts a message to every device on the local network asking, "Who has this IP address?"

ARP Reply: The specific device that owns that IP address sends a direct response back with its physical MAC address.

ARP Cache: The requesting device saves this IP-to-MAC mapping in a temporary storage table called an ARP cache so it does not need to ask again right away.
Why ARP Matters: Bridges Network Layers: It connects network-layer logical addressing (Layer 3 IP) with data-link-layer physical hardware (Layer 2 MAC).
Enables Local Delivery: Hardware frames cannot reach their physical destination on a local network without a known MAC address.
Reduces Traffic: Caching resolved addresses prevents the network from getting flooded with repeated broadcast requests

Ports & Transport Protocols

Explain Ports

Explain Ports

A computer port is a virtual or physical endpoint that allows a device to direct incoming and outgoing data to the correct application or hardware

Physical vs. Network Ports

Physical Ports: Hardware connectors on a device (like USB, HDMI, or Ethernet) used to plug in external peripherals or cables.

Network/Logical Ports: Virtual connection points managed by an operating system's software to sort and route network traffic to specific services

How Network Ports Work

Addresses and Sockets:

An IP address finds the correct device on a network, while a port number identifies the specific program or service running on that device. Together, they form a socket.

Protocols: Ports operate at the transport layer using protocols like TCP (reliable delivery) or UDP (fast delivery).

Categories of Port Numbers

There are 65,535 possible network port numbers, divided into three main ranges:

Well-known Ports (0–1023): Reserved for standard services, such as port 80 for HTTP and port 443 for HTTPS.

Registered Ports (1024–49151): Assigned to specific proprietary applications or vendor services.

Dynamic/Private Ports (49152–65535): Temporarily allocated as ephemeral ports for client connections

TCP vs UDP

TCP vs UDP

TCP is a slow, reliable, and connection-oriented protocol, whereas UDP is a fast, connectionless, and "fire-and-forget" protocol. Both protocols operate at the transport layer of the network model to move data between devices, but they handle delivery guarantees differently

Key Differences

Connection Setup: TCP uses a 3-way handshake (SYN, SYN-ACK, ACK) to connect before sending data. UDP sends data immediately without any setup or connection confirmation.

Reliability: TCP guarantees data delivery, retransmits lost packets, and puts them in the correct order. UDP does not guarantee delivery, meaning packets can go missing, arrive out of order, or duplicate without notice.

Speed and Overhead: TCP is slower with heavy header sizes (20–60 bytes) due to error checking and flow control. UDP is faster with a tiny fixed header size of 8 bytes.

Data Format: TCP treats data as a continuous byte stream. UDP treats data as independent packets or messages.

Explain TCP 3-way handshake

Explain TCP 3-way handshake

The TCP 3-Way Handshake is a three-step process used by devices on a network to set up a reliable connection before sending any data

It acts like a polite greeting between a client (like your computer) and a server (like a website) to make sure both sides are awake, listening, and ready to talk

The Three Steps

Step 1: SYN (Synchronize)

The client sends a special message called a SYN packet to the server.

This packet says: "I want to talk, and I am starting our message count at number X."

Step 2: SYN-ACK (Synchronize-Acknowledge)

The server replies with a SYN-ACK packet.

This packet says: "I hear you! I am ready, and I am starting my message count at number Y. Also, I acknowledge your message X (by saying $X + 1$)"

Step 3: ACK (Acknowledge)

The client sends a final ACK packet back to the server.

This packet says: "Got it! I acknowledge your message Y (by saying $Y + 1$)."

"The connection is now open, and real data can start flowing"

Network Control & Name Resolution

Explain ICMP

Explain ICMP

ICMP stands for Internet Control Message Protocol. It is a network layer protocol used by routers and other devices to send error messages and operational information.

What ICMP Does

Error Reporting: Tells the sender when a data packet fails to reach its destination (such as when a host is unreachable or a timeout occurs).

Network Diagnostics: Powers common troubleshooting tools like ping (which tests if a device is online) and traceroute (which maps the path packets take).

Traffic Management: Alerts senders if a packet is too large to forward or if a better network route is available.

Key Characteristics

Layer 3 Protocol: Operates at the network layer of the TCP/IP model.

No User Data: Does not carry actual user files or website content; it only delivers control and status messages.

Connectionless: Does not set up a formal connection or handshake before sending a message (unlike TCP)

Explain DNS

Explain DNS

The Domain Name System (DNS) is the phonebook of the internet that translates human-friendly website names into machine-readable IP addresses

How DNS Works

Computers find each other on a network using numbers called IP addresses (like 192.0.2.1). Because numbers are hard to remember, DNS matches names like google.com to those numbers

When you type a web address into your browser, the lookup goes through several steps:

Local Cache Check: Your device checks if it already remembers the IP address from a recent visit

Recursive Resolver: If not found, your request goes to a DNS resolver, usually managed by your internet provider or a public service like Cloudflare

Root Server: The resolver asks a root server, which acts as the top layer of the DNS directory

TLD Server: The root server points the resolver to a Top-Level Domain (TLD) server, such as .com or .org

Authoritative Server: The TLD server directs the request to the domain's authoritative server, which holds the exact IP address

Connection: The IP address goes back to your device, and your browser loads the page

Main DNS Record Types

DNS servers store different types of records to handle various tasks

A Record: Connects a name to an IPv4 address.

AAAA Record: Connects a name to an IPv6 address.

CNAME Record: Points one domain name as an alias to another.

MX Record: Directs email to the correct mail server

Explain DHCP / DORA

Explain DHCP/DORA

The DORA process is a four-step communication method used by the Dynamic Host Configuration Protocol to automatically assign an IP address and network settings to a device joining a network

The Four Steps of DORA

Discover: The client device broadcasts a message (DHCPDISCOVER) onto the local network to find any available DHCP server.

Offer: The DHCP server responds with an available IP address, subnet mask, default gateway, and lease time in a DHCPOFFER message.

Request: The client replies with a DHCPREQUEST message to formally accept the offered IP address from that specific server.

Acknowledge: The server sends a final DHCPACK confirmation message, completing the lease and allowing the client to start using the IP address

Web Protocols

Explain HTTP vs HTTPS

Explain HTTP vs HTTPS

The primary difference between HTTP and HTTPS is security. HTTP (HyperText Transfer Protocol) transmits data over the internet in plain text, making it vulnerable to interception. HTTPS (HyperText Transfer Protocol Secure) is the modern, secure version of HTTP that encrypts data using SSL/TLS (Secure Sockets Layer/Transport Layer Security) certificates to ensure privacy and data integrity

HTTP Port no. 80 and HTTPS port no. 443

Key Technical Concepts

Data Encryption: Under HTTP, anyone sitting between your browser and the web server (like a hacker on public Wi-Fi or an ISP) can read transmitted passwords, messages, and credit card numbers. HTTPS scrambles this data into unreadable code before transmission, rendering intercepted data useless

Authentication via Certificates: To use HTTPS, a website operator must obtain an SSL/TLS certificate from a trusted Certificate Authority (CA). This certificate verifies the identity of the server, preventing attackers from creating a fake version of a legitimate website

The Handshake: Before exchanging webpage data, an HTTPS connection establishes a Secure Connection Process. The browser verifies the website's certificate, and the two systems create unique session keys to encrypt all subsequent traffic

Common Ports You Should Know

Port	Common protocol/service
20/21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
143	IMAP
443	HTTPS
445	SMB
3306	MySQL
3389	RDP
5432	PostgreSQL

Note: This PDF is a formatted version of the Day 02 notes provided by the learner. The wording and scope have been preserved rather than silently replacing the original content.